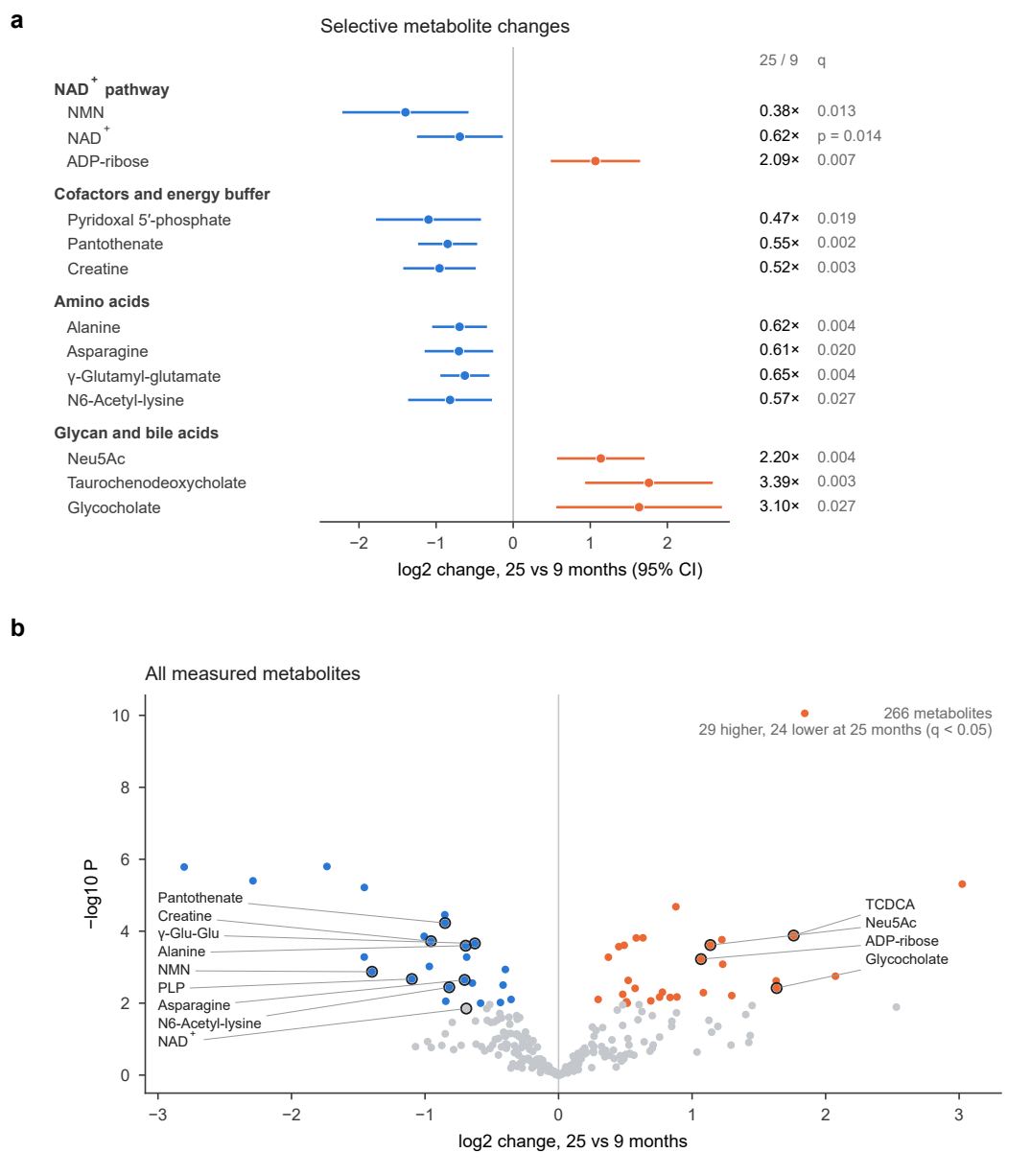

a
Selective metabolite changes
25 / 9
q
NAD+ pathway
NMN
0.38×
0.013
NAD+
0.62×
p = 0.014
ADP-ribose
2.09×
0.007
Cofactors and energy buffer
Pyridoxal 5′-phosphate
0.47×
0.019
Pantothenate
0.55×
0.002
Creatine
0.52×
0.003
Amino acids
Alanine
0.62×
0.004
Asparagine
0.61×
0.020
γ-Glutamyl-glutamate
0.65×
0.004
N6-Acetyl-lysine
0.57×
0.027
Glycan and bile acids
Neu5Ac
2.20×
0.004
Taurochenodeoxycholate
3.39×
0.003
Glycocholate
3.10×
0.027
−2
−1
0
1
2
log2 change, 25 vs 9 months (95% CI)
b
All measured metabolites
266 metabolites
29 higher, 24 lower at 25 months (q < 0.05)
−log10 P
10
8
6
4
2
0
Pantothenate
Creatine
γ-Glu-Glu
Alanine
NMN
PLP
Asparagine
N6-Acetyl-lysine
NAD+
TCDCA
Neu5Ac
ADP-ribose
Glycocholate
−3
−2
−1
0
1
2
3
log2 change, 25 vs 9 months